

FVM / TensorLBM matched Taylor-Green comparison

Same $L=2\pi$ m, $\rho=1$ kg/m³, $\nu=0.1$ m²/s, $U_0=1$ m/s, final time 0.5 s, CPU single thread, float64 states. FVM uses actual MAC face locations and four replicated z planes; LBM uses D2Q9 node locations. Pressure errors use the correct individual midpoint/end clocks.

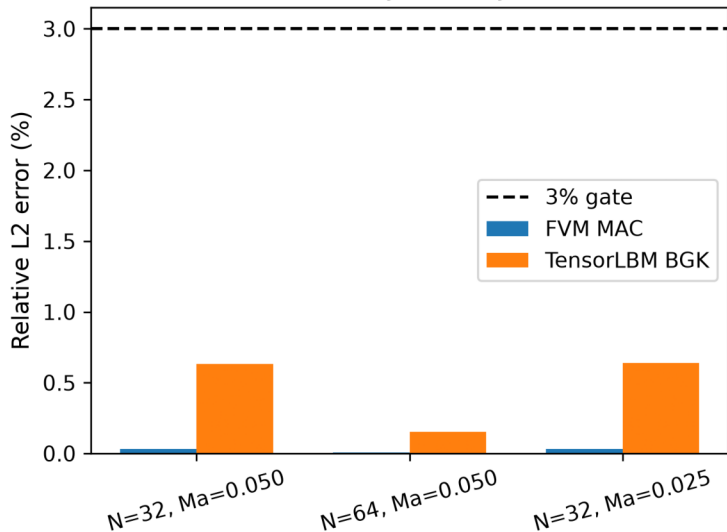
FVM strengths: incompressible pressure constraint, discrete face continuity, stepwise momentum/work/energy audit, physical time step independent of lattice Mach scaling. Primary state storage per physical plane is $4N^2$ doubles versus $9N^2$ D2Q9 populations. This is not a peak-memory result.

LBM is run with unmodified production equilibrium, BGK collision and stream modules. Upstream weights retain float32 constants in float64 populations. Mach sensitivity and population histories are retained. This compares one BGK baseline, not all TensorLBM capabilities.

Observed single-run CPU times include diagnostics and sampling. Different z replication, time steps and solver setup prevent a speed superiority claim. LBM local updates avoid global pressure solves. No GPU or peak RSS claim is made.

N	LBM Ma	FVM u %	LBM u %	FVM p %	LBM p %
32	0.0496	0.0321	0.6335	0.8978	0.8617
64	0.0498	0.0080	0.1521	0.2248	0.8143
32	0.0249	0.0321	0.6400	0.8978	1.6831

velocity accuracy



pressure accuracy

